

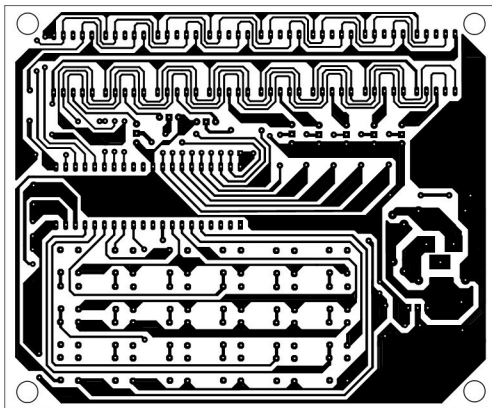


Fig. 2: An actual-size, single-side PCB for the electronic calculator

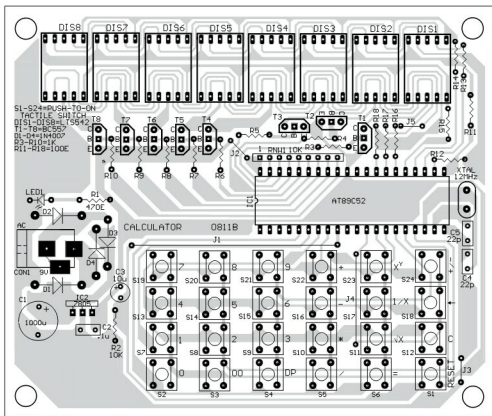


Fig. 3: Component layout for the PCB

Number 1 '*' Number 2 '='

The answer will be displayed

Division.

Dividend '/' divisor '='

The answer is the quotient

Reciprocal.

Number '1/X'

The answer is the inverse of X

Square root.

Number '√X'

The answer is the square root of X

Power.

Number X 'X^Y' Number Y '='

The answer will be X raised to power Y

To toggle +/- sign

Press '+/-'

The sign of the number will be toggled. (The negative sign '-' is displayed on DIS8.)

To erase the DIS1 digit from the displayed number.

Number '←'

Suppose the displayed number = 102.053, then after pressing '←' the display shows number = 102.05

To cancel the displayed number.

Press 'C'. The displayed number is erased and the display shows 0.

To clear all and start new calculation.

Press 'Reset'

Author's prototype is shown in Fig. 4.

Software

The program is written in 'C' language and compiled using Keil μ Vision4 compiler. Keil μ Vision4 is freely available on <https://www.keil.com/c51/demo/eval/c51.htm> website.

The header files AT89X52.h, intrins.h and math.h are included to use the inbuilt functions of the